

DATA ANALYST - SECURITISATION

AI AGENT PROJECT REPORT

Auto Loan Securitisation Risk Assessment Dashboards

Using Advanced DAX Analytics in Microsoft Power BI

Project Code:	483553A
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Target:	Undergraduate / MBA / CA (Inter)
Platform:	Microsoft Power BI DAX Studio Excel

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1. EXECUTIVE SUMMARY

This project presents the development of a comprehensive **Securitisation Risk Assessment Dashboard Suite** built using Microsoft Power BI and advanced Data Analysis Expressions (DAX). The project is designed as part of the ZeTheta AOD (Add Experience to Knowledge) program, targeting undergraduate and MBA students seeking expertise in structured finance analytics. The dashboard suite covers the full spectrum of securitisation risk monitoring for Auto Loan Asset-Backed Securities (ABS), incorporating real-world data requirements from major rating agencies including S&P; Global, Moody's, Fitch, CRISIL, and ICRA.

The core challenge addressed in this project is: *How does a data analyst transform raw loan-level data from SAP/ERP systems into the comprehensive suite of risk metrics that credit committees, rating agencies, investors, and regulators demand?* The answer lies in mastering advanced DAX patterns—CALCULATE with complex filter modifications, iterator functions for weighted computations, virtual tables for dynamic stratification, time intelligence for trend analysis, and context transition for portfolio-level aggregation—combined with deep domain knowledge of securitisation structures, waterfall mechanics, IFRS 9 ECL computation, and regulatory reporting.

1.1 Key Achievements

- **Five Interactive Dashboards:** Executive Overview, DPD/Delinquency Analysis, Dynamic Loss Analysis, Static Pool/Vintage Analysis, and Risk Monitoring Dashboard
- **80+ Production-Grade DAX Measures:** Covering pool health, weighted averages, concentration metrics, time intelligence, and IFRS 9 ECL computation
- **Star Schema Data Model:** Optimized dimensional model with 5 dimension tables and 3 fact tables supporting 500+ loan records
- **Comprehensive Risk Metrics:** DPD rates, CDR, CPR, Recovery Rate, Excess Spread, Pool Factor, and Static CNL Rate
- **Regulatory Compliance Framework:** Alignment with RBI Revised Securitisation Framework (2021) and SEBI disclosure requirements
- **Gamified Simulation Concept:** Securitisation Risk Arena—a browser-based training platform with 8 progressive levels and 1000-point scoring framework

1.2 Portfolio Overview

Metric	Value	Significance
Total Current Balance	■317.79 Million	Aggregate outstanding principal across all active loans
Pool Factor	0.58	Ratio of current balance to original balance—indicates pool amortization
Delinquency Rate (30+ DPD)	0.05%	Percentage of loans 30+ days past due—key risk indicator
Serious Delinquency (60+ DPD)	0.03%	Percentage of loans 60+ days past due—early warning signal

Metric	Value	Significance
Default Rate (90+ DPD)	0.02%	Percentage of loans in default per RBI NPA classification
Weighted Average Coupon (WAC)	11.01%	Interest rate weighted by current balance
Constant Default Rate (CDR)	0.04%	Annualized monthly default rate—standardized loss metric
Conditional Prepayment Rate (CPR)	0.09%	Annualized voluntary prepayment speed
Recovery Rate	0.36%	Percentage of gross losses recovered through collateral liquidation
Excess Spread	■ 216.53 Million	Net interest income available to absorb losses

Note: The portfolio demonstrates strong credit quality with minimal delinquency and robust excess spread coverage. The low CDR and CPR indicate stable borrower behavior with limited prepayment activity. The recovery rate of 36% aligns with industry benchmarks for Indian auto loan ABS, reflecting effective collateral management.

2. PROJECT OVERVIEW & OBJECTIVES

The **Securitisation AI Agent** project is an enterprise-scale financial analytics initiative where data analyst interns build sophisticated risk assessment dashboards for securitised asset portfolios. Working with assumed data extracted from SAP/ERP systems of a financial institution, participants develop complex DAX formulations used across Microsoft data platforms including Power BI, Analysis Services, and Power Pivot.

2.1 Securitisation Fundamentals

Securitisation is the process of pooling various types of financial assets (loans, receivables, mortgages) and converting them into tradeable securities backed by the cash flows from those assets. The process involves transferring assets from an originator to a Special Purpose Vehicle (SPV), which issues securities (tranches) with varying risk-return profiles to investors.

Key Structures:

- **Pass-Through Securities:** Cash flows from underlying assets pass directly to investors proportionally
- **Pay-Through Securities (CMOs):** Cash flows are restructured into sequential tranches with different maturities
- **Credit-Tranched Securities (CDOs/CLOs):** Cash flows allocated by seniority—senior tranches absorb losses last, equity tranches absorb first

2.2 DAX Fundamentals for Securitisation

Every DAX expression evaluates within a context. Understanding context is the single most critical skill for securitisation analytics. The three types of context are:

Context Type	Description	Example Application
Filter Context	Set of filters applied to data before calculation	Total Outstanding Balance = SUM(Loans[CurrentBalance])
Row Context	Iterator that evaluates expression for each row	Weighted Average LTV using SUMX iterator
Context Transition	Converting row context to filter context	Portfolio ECL = SUMX(Loans, CALCULATE([PD]*[LGD]*Loans[EAD]))

2.3 Project Scope and Deliverables

The project spans six comprehensive parts, each building upon the previous to create a complete securitisation analytics framework. The following table outlines the project structure and associated deliverables:

Part	Title	Key Deliverables	Weight
Part A	Training Material & Learning Resources	Securitisation fundamentals, DAX analytics, data requirements across asset classes	20%
Part B	Gamified Simulation Business Concept	Securitisation Risk Arena design, 1000-point scoring framework, 5-year revenue projection	15%
Part C	Case Studies & Real-World Applications	2008 GFC analysis, IL&FS; crisis, RBI framework impact assessment	15%
Part D	15-Day Project Methodology	Day-by-day development plan, 80+ DAX measures, 8+ dashboards	25%
Part E	No-Code Tools & Platforms Guide	Power BI, DAX Studio, Excel, Figma, regulatory data sources	10%
Part F	Submission Protocol	1000-point assessment criteria, validation checklist, IP protection	15%

2.4 Technology Stack

Category	Technology	Purpose
Primary Analytics	Microsoft Power BI Desktop	Dashboard development, DAX authoring, data modeling
DAX Debugging	DAX Studio	Query profiling, performance testing (<100ms target)
Data Modeling	Tabular Editor	Advanced model editing, best practice rules
Data Source	Auto Loan CSV (500 loans, 50+ fields)	Loan-level performance data from SAP/ERP systems
Spreadsheet	Microsoft Excel / Power Pivot	Prototype measures, cross-validation, ECL verification
Version Control	GitHub (Private Repository)	Code versioning, documentation, collaboration
AI Assistance	Claude.ai, ChatGPT, Cursor	DAX generation, research synthesis, concept validation
Design	Figma / Canva / draw.io	Dashboard wireframes, ER diagrams, presentation design

3. DATA ARCHITECTURE & STAR SCHEMA

The foundation of any robust securitisation analytics platform is a well-designed data architecture. This project implements a **star schema** data model—a dimensional modeling approach optimized for analytical queries and reporting performance. The star schema separates descriptive attributes (dimensions) from quantitative measures (facts), enabling efficient DAX computations and intuitive dashboard design.

3.1 Data Model Design

The star schema consists of **5 dimension tables** and **3 fact tables**, connected through foreign key relationships. This design ensures data integrity, supports complex analytical queries, and enables the 80+ DAX measures required for comprehensive securitisation risk assessment. The model is built to handle 500+ loan records with 50+ fields per loan, spanning origination data, performance history, and static pool vintage information.

Star Schema Architecture:

- **Central Fact Table:** FactLoanPerformance connects all dimensions and stores monthly performance metrics
- **Dimension Tables:** DimLoan, DimDate, DimGeography, DimBorrower, DimVehicle provide descriptive context
- **Auxiliary Fact Tables:** FactStaticPool and FactDynamicLoss support vintage analysis and point-in-time reporting
- **Role-Playing Dimensions:** DimDate serves multiple roles (Origination Date, Reporting Date, Snapshot Date)

3.2 Dimension Tables

Dimension Table	Key Fields	Business Purpose	Record Count
DimLoan	LoanID, PoolID, OriginatorID, ServicerID, AccountStatus	Loan identification and origination tracking	500+
DimDate	Date, FiscalYear, FiscalQuarter, Month, MonthEndFlag	Time intelligence and period comparisons	36 months
DimGeography	Region, State, City, PINCode, MetroClassification	Geographic concentration and risk segmentation	25+ regions
DimBorrower	BorrowerID, CIBILScore, EmploymentType, AnnualIncome, DTI	Borrower credit quality and affordability assessment	500+
DimVehicle	VehicleMake, VehicleModel, VehicleYear, VehicleType, FuelType	Collateral valuation and residual value analysis	15+ makes

3.3 Fact Tables

Fact Table	Key Measures	Granularity	Update Frequency
FactLoanPerformance	CurrentBalance, DPD_Days, DPD_Bucket, AmountOverdue, EMIOverdue	Loan-Month	Monthly
FactStaticPool	CumulativeDefaults, CumulativeGrossLoss, CumulativeRecoveries, CumulativeNetLoss, PoolFactor	Vintage-Month	Monthly
FactDynamicLoss	BOP_Balance, EOP_Balance, NewDefaults, GrossLoss, NetLoss, Prepayments, Collections	Pool-Month	Monthly

3.4 Relationship Mapping

The star schema relationships are configured to support both **single-direction** and **bi-directional** filtering, depending on the analytical requirement. The primary relationships are:

From Table	To Table	Relationship Type	Filter Direction
DimLoan	FactLoanPerformance	One-to-Many	Single (DimLoan → Fact)
DimDate	FactLoanPerformance	One-to-Many	Single (DimDate → Fact)
DimBorrower	FactLoanPerformance	One-to-Many	Single (DimBorrower → Fact)
DimVehicle	FactLoanPerformance	One-to-Many	Single (DimVehicle → Fact)
DimGeography	FactLoanPerformance	One-to-Many	Single (DimGeography → Fact)
DimDate (Role-Playing)	FactStaticPool	One-to-Many	Single (VintageDate → Fact)
DimDate (Role-Playing)	FactDynamicLoss	One-to-Many	Single (ReportingDate → Fact)

Role-Playing Dimensions: The DimDate table is used multiple times in the model to support different date contexts. For example, the same DimDate table serves as the Origination Date for vintage analysis and the Reporting Date for dynamic loss analysis. This is achieved through inactive relationships that are activated within specific DAX measures using the USERELATIONSHIP function.

4. DAX MEASURE LIBRARY

The DAX measure library forms the computational backbone of the securitisation analytics platform. This section documents the 80+ production-grade measures developed for the project, organized by functional category. Each measure is designed for performance (target <100ms execution time in DAX Studio) and auditability (fully formula-driven with no hard-coded values).

4.1 Pool Health Measures

Pool health measures provide the foundational metrics for portfolio monitoring. These measures compute aggregate balances, delinquency rates, and collection efficiency indicators that are essential for monthly investor reporting.

```
1. Total Outstanding Balance Total Outstanding Balance = SUM(Loans[CurrentBalance]) 2. Pool Factor Pool Factor = DIVIDE( SUM(Loans[CurrentBalance]), CALCULATE(SUM(Loans[OriginalLoanAmount]), ALL(DimDate)) ) 3. Delinquency Rate (30+ DPD) Delinquency Rate = DIVIDE( CALCULATE(SUM(Loans[CurrentBalance]), Loans[DelinquencyDays] >= 30), SUM(Loans[CurrentBalance]) ) 4. Serious Delinquency Rate (60+ DPD) Serious Delinquency Rate = DIVIDE( CALCULATE(SUM(Loans[CurrentBalance]), Loans[DelinquencyDays] >= 60), SUM(Loans[CurrentBalance]) ) 5. Collection Efficiency Collection Efficiency = DIVIDE(SUM(Loans[ActualPayment]), SUM(Loans[ScheduledPaymentDue])) 6. Constant Default Rate (CDR) - Annualised CDR = VAR MonthlyDefault = [Defaults This Month] VAR BOP_Balance = CALCULATE(SUM(Loans[CurrentBalance]), PREVIOUSMONTH(DimDate[Date])) VAR MDR = DIVIDE(MonthlyDefault, BOP_Balance) RETURN 1 - POWER(1 - MDR, 12)
```

4.2 Weighted Average Measures

Weighted average measures are critical for rating agency reporting and investor analysis. All weighted averages use the SUMX iterator pattern to compute balance-weighted metrics, ensuring accurate portfolio-level aggregation.

```
7. Weighted Average Coupon (WAC) WAC = DIVIDE( SUMX(Loans, Loans[InterestRate] * Loans[CurrentBalance]), SUM(Loans[CurrentBalance]) ) 8. Weighted Average Maturity (WAM) WAM = DIVIDE( SUMX(Loans, Loans[RemainingTerm] * Loans[CurrentBalance]), SUM(Loans[CurrentBalance]) ) 9. Weighted Average Loan Age (WALA) WALA = DIVIDE( SUMX(Loans, Loans[MonthsOnBook] * Loans[CurrentBalance]), SUM(Loans[CurrentBalance]) ) 10. Weighted Average LTV Weighted Avg LTV = DIVIDE( SUMX(FactLoanPerformance, FactLoanPerformance[LTV_Current] * FactLoanPerformance[CurrentBalance]), SUM(FactLoanPerformance[CurrentBalance]) ) 11. Weighted Average CIBIL Score Weighted Avg CIBIL = DIVIDE( SUMX(Loans, Loans[CIBIL_Score_Current] * Loans[CurrentBalance]), SUM(Loans[CurrentBalance]) ) 12. Weighted Average DTI Weighted Avg DTI = DIVIDE( SUMX(Loans, Loans[DTI_Ratio] * Loans[CurrentBalance]), SUM(Loans[CurrentBalance]) )
```

4.3 Concentration & Risk Metrics

Concentration metrics measure portfolio diversification and identify potential risk concentrations. The Herfindahl-Hirschman Index (HHI) and Top N Borrower Exposure are standard measures required by rating agencies for assessing portfolio concentration risk.

```
13. Herfindahl-Hirschman Index (HHI) HHI Index = VAR TotalBal =  
CALCULATE(SUM(Loans[CurrentBalance]), ALL(DimBorrower)) VAR BorrowerShares = ADDCOLUMNS(  
SUMMARIZE(Loans, DimBorrower[BorrowerID]), "Share", DIVIDE(CALCULATE(SUM(Loans[CurrentBalance])),  
TotalBal) ) RETURN SUMX(BorrowerShares, [Share] ^ 2) 14. Top 10 Borrower Exposure Top N Borrower  
Exposure = VAR TotalBal = CALCULATE(SUM(Loans[CurrentBalance]), ALL(DimBorrower)) VAR  
BorrowerRanked = ADDCOLUMNS( SUMMARIZE(Loans, DimBorrower[BorrowerID]), "Balance",  
CALCULATE(SUM(Loans[CurrentBalance])) ) VAR TopN = TOPN(10, BorrowerRanked, [Balance], DESC)  
RETURN DIVIDE(SUMX(TopN, [Balance]), TotalBal) 15. Geographic Concentration (Top 5 States) Top 5  
State Concentration = VAR TotalBal = SUM(Loans[CurrentBalance]) VAR StateRanked = ADDCOLUMNS(  
SUMMARIZE(Loans, DimGeography[State]), "Balance", CALCULATE(SUM(Loans[CurrentBalance])) ) VAR  
Top5 = TOPN(5, StateRanked, [Balance], DESC) RETURN DIVIDE(SUMX(Top5, [Balance]), TotalBal) 16.  
Vehicle Make Concentration Vehicle Make HHI = VAR TotalBal = SUM(Loans[CurrentBalance]) VAR  
MakeShares = ADDCOLUMNS( SUMMARIZE(Loans, DimVehicle[VehicleMake]), "Share",  
DIVIDE(CALCULATE(SUM(Loans[CurrentBalance])), TotalBal) ) RETURN SUMX(MakeShares, [Share] ^ 2)
```

4.4 Time Intelligence Measures

Time intelligence measures enable trend analysis, period-over-period comparisons, and rolling averages. These measures are essential for detecting deterioration signals and providing early warning indicators to portfolio managers.

```
17. YTD Delinquency Rate YTD Delinquency Rate = CALCULATE( [Delinquency Rate],  
DATESYTD(DimDate[Date]) ) 18. Rolling 3-Month Average DPD Rate 3M Avg 30Plus DPD Rate = AVERAGEX(  
DATESINPERIOD(DimDate[Date], MAX(DimDate[Date]), -3, MONTH), CALCULATE([30Plus DPD Rate by  
Month]) ) 19. Period-over-Period Change DPD Rate MoM Change = VAR CurrentMonth = [30Plus DPD Rate  
by Month] VAR PriorMonth = CALCULATE([30Plus DPD Rate by Month], DATEADD(DimDate[Date], -1,  
MONTH)) RETURN DIVIDE(CurrentMonth - PriorMonth, PriorMonth) 20. DPD Trend Direction DPD Trend  
Direction = VAR Current3M = [3M Avg 30Plus DPD Rate] VAR Prior3M = CALCULATE([3M Avg 30Plus DPD  
Rate], DATEADD(DimDate[Date], -3, MONTH)) RETURN SWITCH( TRUE(), Current3M > Prior3M * 1.1,  
"Deteriorating", Current3M < Prior3M * 0.9, "Improving", "Stable" )
```

4.5 IFRS 9 ECL Computation

The IFRS 9 Expected Credit Loss (ECL) model is a critical regulatory requirement for financial institutions. The DAX implementation covers all three stages of credit impairment, from 12-month ECL for Stage 1 assets to lifetime ECL for Stage 2 and Stage 3 assets.

```
21. Stage Classification IFRS 9 Stage = SWITCH( TRUE(), Loans[DPD_Days] >= 120 ||  
Loans[IsDefaulted] = TRUE, 3, // Stage 3 Loans[DPD_Days] >= 60 && Loans[DPD_Days] < 120, 2, //  
Stage 2 1 // Stage 1 ) 22. 12-Month ECL (Stage 1) 12M ECL = SUMX( FILTER(Loans, [IFRS 9 Stage] =  
1), CALCULATE([PD_12M] * [LGD] * Loans[EAD]) ) 23. Lifetime ECL (Stage 2 & 3) Lifetime ECL = SUMX(  
FILTER(Loans, [IFRS 9 Stage] IN {2, 3}), CALCULATE([PD_Lifetime] * [LGD] * Loans[EAD]) ) 24. Total  
ECL Provision Total ECL = [12M ECL] + [Lifetime ECL] 25. ECL Coverage Ratio ECL Coverage Ratio =  
DIVIDE([Total ECL], SUM(Loans[CurrentBalance]))
```

Validation Note: All ECL measures are cross-validated against Excel computations with a target variance of <0.01%. The PD term structures are calibrated using historical transition matrices, and LGD estimates

incorporate recovery rate data from the pool's actual collection experience.

5. DASHBOARD ANALYSIS

The securitisation analytics platform comprises five interconnected dashboards, each designed to serve a specific analytical purpose while maintaining visual consistency and interactive functionality. This section provides a detailed analysis of each dashboard, including the visual components, underlying DAX measures, and business insights derived from the data.

5.1 Executive Overview Dashboard

The Executive Overview Dashboard serves as the primary interface for senior management and portfolio managers. It provides a high-level summary of portfolio health, geographic distribution, and key performance indicators in a single consolidated view. The dashboard is designed for quick consumption, with KPI cards prominently displayed and drill-through capabilities enabled for deeper analysis.

Dashboard Components:

- **Total Outstanding Balance (Donut Chart):** Visualizes the aggregate portfolio size (■317.79M) with 100% utilization indicator
- **Total Current Balance (KPI Card):** Primary metric showing ■317.79M with conditional formatting for threshold breaches
- **Pool Factor (KPI Card):** Current ratio of 0.58 indicating 42% amortization since origination
- **Delinquency Rate (KPI Card):** Overall delinquency at 0.14% with trend arrow indicating stability
- **Average Interest Rate (KPI Card):** Weighted average coupon of 11.01% supporting excess spread analysis
- **Total Outstanding by Vehicle Type (Bar Chart):** Segmentation across SUV, MUV, Compact, Sedan, and Hatchback categories
- **Total Outstanding by Region (Bar Chart):** Geographic distribution across East, North, West, and South regions
- **Sum of Current Balance by State (Map):** Interactive map of India with bubble markers indicating state-level concentrations
- **Region/Vintage Slicer:** Multi-select filter enabling dynamic segmentation by region and origination vintage

Key Insights from Executive Overview:

- **Portfolio Scale:** The ■317.79M current balance represents a mid-sized auto loan ABS pool, suitable for institutional investor participation
- **Amortization Progress:** Pool factor of 0.58 indicates significant seasoning (42% paid down), suggesting mature collateral with predictable cash flows
- **Geographic Concentration:** East and North regions dominate the portfolio, requiring monitoring for regional economic downturns
- **Vehicle Mix:** SUVs and MUVs constitute the majority, reflecting premium segment focus with potentially lower default rates but higher severity

- **Interest Rate Environment:** 11.01% WAC provides substantial excess spread cushion (■216.53M) for loss absorption

***Dashboard Screenshot Reference:** Screenshot (16) and Screenshot (17) depict the Executive Overview Dashboard in design mode and view mode respectively. The formatting panel visible in Screenshot (16) indicates active development of visual properties, with background transparency set at 68% and border styling enabled.*

5.2 DPD / Delinquency Analysis Dashboard

The DPD (Days Past Due) Analysis Dashboard is the most critical operational tool for portfolio managers and risk analysts. It provides granular visibility into delinquency migration, bucket distribution, and trigger status monitoring. This dashboard is used daily for early warning detection and monthly for investor reporting compliance.

Dashboard Components:

- **KPI Cards Row:** Current Rate (0.87%), 30+ DPD Rate (0.05%), 60+ DPD Rate (0.03%), 90+ DPD Rate (0.02%) with conditional formatting (green→yellow→red gradient)
- **DPD Distribution by Balance (Stacked Bar Chart):** Horizontal bars showing balance concentration across Current, 1-29 DPD, 30-59 DPD, 60-89 DPD, and 90+ DPD buckets
- **30+ DPD Monthly Trend (Line Chart):** Time-series visualization from Jan 2024 to Sep 2024 showing upward trajectory from near-zero to 0.05%
- **Loan-Level Detail Table:** Tabular view of all delinquent loans with CurrentBalance, DPD_Bucket, DPD_Days, and LoanID for drill-down analysis
- **Delinquency Trigger Status (Gauge Chart):** Semi-circular gauge showing current 30+ DPD rate (0.05) against trigger threshold (0.09), with PASS status indicator

DPD Bucket Definitions (Industry Standard):

DPD Bucket	Days Range	RBI Classification	IFRS 9 Stage	S&P; Treatment
Current	0 DPD	Standard	Stage 1	Performing
1-29 DPD	1-29 days	SMA-0	Stage 1	Performing
30-59 DPD	30-59 days	SMA-1	Stage 2	Delinquent
60-89 DPD	60-89 days	SMA-2	Stage 2	Delinquent
90-119 DPD	90-119 days	Sub-standard	Stage 3	Defaulted
120+ DPD	120+ days	Sub-standard/Loss	Stage 3	Defaulted

Key Insights from DPD Analysis:

- **Low Delinquency Environment:** 30+ DPD rate of 0.05% indicates excellent portfolio health, well below the 5% trigger threshold
- **Trigger Safety Margin:** Current 30+ DPD (0.05) vs trigger threshold (0.09) provides 44% safety margin, indicating robust structural protection
- **Upward Trend Concern:** Monthly trend shows steady increase from Jan 2024 to Sep 2024, requiring close monitoring for acceleration
- **RBI Compliance:** All delinquent loans are classified as SMA-0 or SMA-1, with no NPA exposure (90+ DPD at 0.02%)
- **Loan-Level Granularity:** Table shows specific loans in default (e.g., ZAAUTO2024-1-00494 with 2101 DPD days, █38.7M balance), enabling targeted collection actions

Dashboard Screenshot Reference: Screenshot (18) shows the DPD Analysis Dashboard with all components fully rendered. The DPD_Bucket slicer on the left enables interactive filtering, while the gauge chart provides immediate visual confirmation of trigger status (PASS).

5.3 Dynamic Loss Analysis Dashboard

The Dynamic Loss Analysis Dashboard provides point-in-time portfolio performance metrics essential for monthly investor reporting and cash flow waterfall modeling. Unlike static pool analysis, dynamic analysis measures the current outstanding pool, where the denominator changes monthly as loans prepay, amortize, or default. This dashboard is critical for computing CDR, CPR, excess spread, and trigger compliance.

Dashboard Components:

- **KPI Cards Row:** CDR (0.04%), CPR (0.09%), Recovery Rate (0.36), Monthly Net Loss Rate (0.01), Excess Spread (■216.53M)
- **BOP/EOP Balance Waterfall (Stacked Bar Chart):** Monthly progression showing Beginning Balance, Scheduled Amortization, Prepayments, Defaults, and Ending Balance
- **Monthly Net Loss Rate & CDR Trend (Line Chart):** Dual-axis chart from Jan 2024 to Sep 2024 showing loss rate stability and CDR consistency
- **Collections Breakdown Trend (Area Chart):** Stacked area showing Interest Collections, Principal Collections, Prepayments, and Recoveries over time
- **CDR vs CPR Analysis (Scatter Plot):** Bubble chart comparing default and prepayment speeds by reporting period, with bubble size indicating pool balance
- **Performance Summary Table:** Monthly tabular data with ReportingDate, BOP_Balance, NetLoss, Recoveries, and CollectionEfficiency
- **Delinquency Trigger Status (Gauge):** Current trigger level (0.01) vs threshold (0.03) with PASS indicator

Dynamic Loss Metrics Explained:

Metric	Formula	Current Value	Interpretation
Constant Default Rate (CDR)	Annualized MDR: $1-(1-MDR)^{12}$	0.04%	Low default velocity indicating stable credit quality
Conditional Prepayment Rate (CPR)	Annualized SMM: $1-(1-SMM)^{12}$	0.09%	Minimal voluntary prepayments suggesting borrower stickiness
Recovery Rate	Recoveries / Gross Losses	36%	Below prime auto ABS average (40-60%), indicating room for improvement
Monthly Net Loss Rate	NetLoss / Avg Balance	0.01%	Extremely low loss rate supported by strong excess spread
Excess Spread	Interest - Investor Coupon - Fees - Losses	■216.53M	Substantial cushion providing 999+ months of loss coverage
Collection Efficiency	Actual Collections / Scheduled Collections	97.2%	Strong collection performance with minimal slippage

Key Insights from Dynamic Loss Analysis:

- **Stable Loss Environment:** CDR and CPR both remain below 0.1%, indicating minimal portfolio churn and stable cash flows

- **Excess Spread Strength:** ■216.53M excess spread provides extraordinary protection—equivalent to 999+ months of current monthly losses
- **Collection Performance:** 97.2% collection efficiency demonstrates effective servicing operations with minimal billing slippage
- **Recovery Opportunity:** 36% recovery rate is below industry benchmarks, suggesting potential for enhanced collateral liquidation strategies
- **Waterfall Health:** All trigger tests (delinquency, loss, OC/IC) show PASS status, indicating no structural concerns

Dashboard Screenshot Reference: Screenshot (19) captures the Dynamic Loss Dashboard in full view. The Collections Breakdown Trend shows a peak in May 2024, while the CDR vs CPR scatter plot indicates consistent performance across all reporting periods.

5.4 Static Pool / Vintage Analysis Dashboard

The Static Pool (Vintage) Analysis Dashboard is the gold standard methodology used by rating agencies to project expected losses for securitised pools. Unlike dynamic analysis, static pool analysis tracks a fixed group of loans from origination to termination, eliminating the distorting effects of pool growth, seasoning mix changes, and portfolio management activities. This dashboard is essential for S&P; LEVELS model inputs, Moody's Idealised Default Table calibration, and CRISIL/ICRA surveillance.

Dashboard Components:

- **KPI Cards Row:** Static CNL Rate (0.00), PoolFactor (254.28), RemainingPoolBalance (■8.98bn), 30+ Delinquency Rate (0.03)
- **Cumulative Net Loss Trend (Multi-Line Chart):** Loss curves by vintage quarter (2021-Q1 through 2022-Q4) showing divergence patterns
- **Vintage Loss Curve Analysis (Multi-Line Chart):** Normalized cumulative loss curves enabling cross-vintage comparison at same seasoning
- **Vintage Performance Summary Table:** Tabular data with CurrentDelinq30Plus, CumulativeNetLoss, MonthsOnBook, and RemainingPoolBalance by vintage
- **Vintage Slicer:** Multi-select filter for 2021-Q1 through 2023-Q1 vintages enabling focused analysis
- **MonthsOnBook Slicer:** Seasoning filter (0-7 months) for cohort-specific analysis
- **VintageStartDate Slicer:** Calendar filter for precise origination period selection

Static Pool Data Structure:

Column	Data Type	Description	Example
VintageID	VARCHAR(10)	Origination cohort identifier	2022-Q1
OriginalLoanCount	INT	Loans in static pool at Month 0	1,250
OriginalPoolBalance	DECIMAL(15,2)	Total balance at Month 0	■125,00,00,000
MonthsOnBook	INT	Months since pool inception	24
CumulativeNetLoss	DECIMAL(15,2)	Net loss since Month 0	■1,45,00,000
CumulativeNetLossRate	DECIMAL(8,6)	Net Loss / Original Pool Balance	0.011600
PoolFactor	DECIMAL(5,4)	Remaining / Original balance	0.7000
CurrentDelinq30Plus	DECIMAL(8,6)	30+ DPD rate of surviving loans	0.035000

Key DAX Measures for Static Pool Analysis:

```
Static CNL Rate: VAR SelectedVintage = SELECTEDVALUE(StaticPool[VintageID]) VAR MaxMOB = MAX(StaticPool[MonthsOnBook]) VAR CumNetLoss = CALCULATE(MAX(StaticPool[CumulativeNetLoss]), StaticPool[VintageID] = SelectedVintage, StaticPool[MonthsOnBook] = MaxMOB) VAR OrigBalance = CALCULATE(MAX(StaticPool[OriginalPoolBalance]), StaticPool[VintageID] = SelectedVintage) RETURN DIVIDE(CumNetLoss, OrigBalance) Projected Ultimate CNL (Extrapolation): VAR CurrentCNL = [Static CNL Rate] VAR CurrentMOB = MAX(StaticPool[MonthsOnBook]) VAR ExpectedLifeMonths = 60 VAR LossCurveCompletion = CALCULATE(MAX(LossTimingCurve[CumulativePctOfLoss]), LossTimingCurve[MonthsOnBook] = CurrentMOB) RETURN DIVIDE(CurrentCNL, LossCurveCompletion) Loss
```

Curve Acceleration: VAR CurrentMarginal = [Marginal Monthly Loss Rate] VAR PriorMarginal = CALCULATE([Marginal Monthly Loss Rate], StaticPool[MonthsOnBook] = MAX(StaticPool[MonthsOnBook]) - 1) RETURN IF(CurrentMarginal > PriorMarginal * 1.2, "Accelerating", IF(CurrentMarginal < PriorMarginal * 0.8, "Decelerating", "Stable"))

Key Insights from Static Pool Analysis:

- **Zero Static CNL:** Current cumulative net loss rate of 0.00% indicates pristine vintage performance with no realized losses
- **High Pool Factor:** Value of 254.28 suggests data aggregation across multiple vintages or measurement in different units
- **Vintage Comparison:** 2021-Q1 vintage shows highest cumulative loss (■94.08M), while 2022-Q4 shows lowest (■33.43M), indicating improving underwriting quality
- **Seasoning Pattern:** Loss curves peak around months 18-30 (typical for Indian auto loans), with 2022-Q3 vintage approaching peak loss period
- **Rating Agency Alignment:** Loss timing curve data supports S&P; LEVELS model inputs and Moody's Idealised Default Table calibration

Dashboard Screenshot Reference: Screenshot (20) displays the Static Pool Dashboard with vintage slicers active. The Cumulative Net Loss Trend shows clear divergence between 2021 and 2022 vintages, while the Vintage Performance Summary table provides granular data for rating agency reporting.

5.5 Risk Monitoring Dashboard

The Risk Monitoring Dashboard serves as the centralized command center for real-time portfolio surveillance. It integrates delinquency data, regional risk factors, loan-level detail, and trigger monitoring into a single comprehensive view. This dashboard is designed for daily use by risk analysts and for emergency response during crisis scenarios.

Dashboard Components:

- **KPI Cards Row:** Current Rate (--), 30+ DPD % (1.00), 60+ DPD % (1.00), 90+ DPD % (1.00) with color-coded backgrounds (red→yellow→blue→green)
- **DPD by Region Chart (Line Chart):** Geographic delinquency trend showing South, North, West, East, and Central regions with balance-weighted values
- **Monthly Risk Trend (Line Chart):** Multi-line chart tracking Current Rate, 90+ DPD Rate, 30+ DPD Rate, and 60+ DPD Rate from Jul 2024 to Oct 2024
- **CurrentBalance by LoanID and DPD_Days (Scatter Plot):** Bubble chart showing individual loan positions with DPD days on x-axis and balance on y-axis
- **DPD_Bucket Slicer (Hierarchical):** Expandable tree showing 120+ DPD, 1-29 DPD, 30-59 DPD, 60-89 DPD, 90-119 DPD, Current, Default, Repossessed
- **Risk Detail Table:** Loan-level data with SnapshotDate, LoanID, and Sum of CurrentBalance for audit trail
- **Trigger Gauge:** Semi-circular gauge showing current trigger level (1.00) against threshold range (0.00-2.00)

Risk Monitoring Data Structure:

Column	Data Type	Description	Business Use
SnapshotDate	DATE	Reporting period end date	Time-series tracking and trend analysis
LoanID	VARCHAR(20)	Unique loan identifier	De-duplication and drill-down tracking
DPD_Days	INT	Exact number of days past due	Precise delinquency measurement
DPD_Bucket	VARCHAR(15)	Bucket classification	Standardized reporting and trigger testing
CurrentBalance	DECIMAL(15,2)	Balance at snapshot	Pool factor and exposure calculation
Region	VARCHAR(20)	Geographic region	Concentration risk monitoring
RollFlag	VARCHAR(10)	Roll direction (Worse/Better/Same)	Migration trend analysis

Key Insights from Risk Monitoring:

- **Placeholder Values Detected:** KPI cards showing 1.00 for all DPD rates and '--' for Current Rate indicate incomplete measure development or data connectivity issues

- **Regional Risk Distribution:** DPD by Region chart shows declining trend from South to Central, suggesting geographic concentration risk in southern markets
- **Monthly Stability:** Risk trend from Jul 2024 to Oct 2024 shows flat lines, indicating stable but potentially stagnant portfolio performance
- **Loan-Level Granularity:** Risk Detail Table provides audit trail with specific loan identifiers (ZAAUTO2024-1-00010, ZAAUTO2024-1-00046) for regulatory compliance
- **Trigger Status:** Gauge shows 1.00 within acceptable range (0.00-2.00), but precise threshold definitions require clarification

***Dashboard Screenshot Reference:** Screenshot (21) shows the Risk Monitoring Dashboard with all five tabs visible at the bottom: Executive Overview, DPD Analysis, Dynamic Loss, Static Pool, and Risk Monitoring. The DPD_Bucket slicer uses a hierarchical tree structure for intuitive navigation.*

Development Status Note: Several KPI cards in the Risk Monitoring Dashboard display placeholder values (1.00) or incomplete data ('--'), indicating that DAX measures require further development and validation. This is consistent with the 15-day project timeline where Day 7 focuses on DPD analysis and Day 8 on vintage analysis, with risk monitoring scheduled for later phases.

6. SECURITISATION ANALYTICS FRAMEWORK

The securitisation analytics framework provides the methodological foundation for all risk assessment activities. This section details the four core analytical pillars: DPD Analysis, Static Pool/Vintage Analysis, Dynamic Loss Analysis, and Roll Rate Analysis. Each pillar is supported by comprehensive DAX measures, standardized data structures, and regulatory compliance requirements.

6.1 Days Past Due (DPD) Analysis

DPD analysis is the single most scrutinized section of any securitisation performance report. Rating agencies, investors, trustees, and regulators all monitor DPD tables monthly to assess portfolio health, predict future losses, and determine whether structural triggers have been breached. The DAX model implements industry-standard bucket definitions with flexibility for jurisdiction-specific requirements.

6.1.1 DPD Bucket Definitions and Standards

The industry-standard DPD buckets used across all securitisation asset classes are defined below. The DAX model accommodates variations across RBI, S&P, Moody's, and IFRS 9 classification systems:

DPD Bucket	Days Range	RBI Classification	IFRS 9 Stage	S&P; Treatment	Moody's Treatment
Current	0 DPD	Standard	Stage 1	Performing	Performing
1-29 DPD	1-29 days	SMA-0	Stage 1	Performing	Performing
30-59 DPD	30-59 days	SMA-1	Stage 2	Delinquent	Delinquent
60-89 DPD	60-89 days	SMA-2	Stage 2	Delinquent	Delinquent
90-119 DPD	90-119 days	Sub-standard	Stage 3	Defaulted	Defaulted
120-149 DPD	120-149 days	Sub-standard	Stage 3	Defaulted	Defaulted
150-179 DPD	150-179 days	Doubtful	Stage 3	Defaulted	Defaulted
180-359 DPD	180-359 days	Doubtful	Stage 3	Defaulted	Charged-off
360+ DPD	360+ days	Loss	Stage 3	Loss	Loss

6.1.2 RBI Special Mention Account (SMA) Classification

The Reserve Bank of India requires banks to report SMA classifications as an early warning system. For securitised pools, these classifications feed directly into rating agency surveillance. Under RBI's revised framework (November 2021), the date of NPA classification is determined by the exact day count, not by the end of the month. This affects how DAX measures count delinquent loans:

SMA Category	Overdue Period	Action Required	DAX Measure Impact
SMA-0	Principal/interest overdue 1-30 days	Strengthen monitoring	DPD_Days >= 1 && DPD_Days <= 30
SMA-1	Principal/interest overdue 31-60 days	Initiate corrective action	DPD_Days >= 31 && DPD_Days <= 60
SMA-2	Principal/interest overdue 61-90 days	Escalate to recovery team	DPD_Days >= 61 && DPD_Days <= 90
NPA	Principal/interest overdue >90 days	Classify as non-performing	DPD_Days > 90

6.1.3 DPD Distribution Report - Core DAX Measures

The following DAX measures generate the complete DPD distribution report required by investors. Each measure produces both loan-count and balance-weighted views:

Measure Group 1: DPD Bucket Counts and Balances DPD Loan Count = COUNTROWS(FactLoanPerformance)
 DPD Balance = SUM(FactLoanPerformance[CurrentBalance]) DPD Percentage of Pool (by Balance) =
 DIVIDE([DPD Balance], CALCULATE([DPD Balance], REMOVEFILTERS(FactLoanPerformance[DPD_Bucket])))
 DPD Percentage of Pool (by Count) = DIVIDE([DPD Loan Count], CALCULATE([DPD Loan Count],
 REMOVEFILTERS(FactLoanPerformance[DPD_Bucket]))) **Measure Group 2: Specific Bucket Measures (for KPI Cards)** Current Balance = CALCULATE(SUM(FactLoanPerformance[CurrentBalance]),
 FactLoanPerformance[DPD_Days] = 0) 30Plus DPD Balance =
 CALCULATE(SUM(FactLoanPerformance[CurrentBalance]), FactLoanPerformance[DPD_Days] >= 30) 30Plus
 DPD Rate = DIVIDE([30Plus DPD Balance], SUM(FactLoanPerformance[CurrentBalance])) 60Plus DPD
 Balance = CALCULATE(SUM(FactLoanPerformance[CurrentBalance]), FactLoanPerformance[DPD_Days] >=
 60) 90Plus DPD Balance = CALCULATE(SUM(FactLoanPerformance[CurrentBalance]),
 FactLoanPerformance[DPD_Days] >= 90) NPA Rate (RBI) = DIVIDE([90Plus DPD Balance],
 SUM(FactLoanPerformance[CurrentBalance]))

6.2 Static Pool / Vintage Analysis

Static pool analysis (also called vintage analysis or static loss analysis) is the gold standard methodology used by rating agencies to project expected losses for a securitised pool. Unlike dynamic analysis which looks at the current snapshot, static pool analysis tracks a fixed group of loans from origination to termination, regardless of when new loans are added or existing loans pay off. This pure measurement of credit performance eliminates the distorting effects of pool growth, seasoning mix changes, and portfolio management.

6.2.1 Why Rating Agencies Require Static Loss Data

- **Eliminates Denominator Effect:** New originations don't dilute delinquency rates, providing pure credit performance measurement
- **Pure Credit Performance:** Measures only credit deterioration, not portfolio management activities like originations or prepayments
- **Vintage Comparison:** Enables direct comparison across origination cohorts to identify underwriting quality trends
- **Loss Curve Projection:** Allows extrapolation of partial curves to project ultimate losses for incomplete vintages
- **Originator Assessment:** Evaluates whether underwriting quality is stable, improving, or deteriorating over time

6.2.2 Loss Timing Curve (S&P;/Moody's Requirement)

The loss timing curve defines what percentage of total expected lifetime losses occur in each month. Rating agencies use this to time-stress cash flows. The standard loss timing curve for Indian auto loans peaks around months 18-30:

MOB Range	% of Total Loss	Cumulative %	Typical Auto	Typical RMBS	Typical Cards	Typical CLO
0-6	5%	5%	Low	Very Low	Moderate	Very Low
7-12	12%	17%	Rising	Low	Moderate	Low
13-18	18%	35%	Peak	Rising	Moderate	Rising
19-24	22%	57%	Peak	Rising	Moderate	Moderate
25-36	25%	82%	Declining	Peak	Moderate	Peak
37-48	12%	94%	Low	Declining	Declining	Declining
49-60	6%	100%	Tail	Declining	Low	Tail

6.2.3 Static vs Dynamic Loss Comparison

Dimension	Static Loss Data	Dynamic Loss Data
Pool Definition	Fixed cohort at origination	Current outstanding pool
Denominator	Original pool balance (fixed)	Current pool balance (changes monthly)
Time Horizon	From origination to final payment	Current reporting period
New Originations	Excluded after cutoff	Included in pool
Prepayments	Tracked but pool stays fixed	Reduce the current denominator
Primary Use	Rating agency loss projection	Monthly investor reporting
Key Metric	Cumulative Net Loss Rate	Annualised Default/Loss Rate
Distortion Risk	None (pure measurement)	Denominator effect from pool changes
DAX Complexity	Moderate (time-series)	High (point-in-time with prior period references)

6.3 Dynamic Loss Analysis

Dynamic loss data measures the performance of the current pool at a specific point in time. Unlike static pool analysis, the denominator (pool size) changes every month as loans prepay, amortise, default, or are added. Dynamic analysis is used for monthly investor reporting, trigger testing, and operational portfolio management. The DAX implementation requires sophisticated time intelligence and prior-period referencing.

6.3.1 Dynamic Loss Data Structure

Column	Data Type	Description	Example
ReportingDate	DATE	End of reporting period	2024-10-31
BOP_LoanCount	INT	Loan count at start of month	475
BOP_Balance	DECIMAL(15,2)	Pool balance at start of month	■32,50,00,000
NewDefaults_Count	INT	Loans defaulting this month	3
NewDefaults_Balance	DECIMAL(15,2)	Balance of new defaults	■18,50,000
GrossLoss_ThisMonth	DECIMAL(15,2)	Gross losses realised	■12,00,000
Recoveries_ThisMonth	DECIMAL(15,2)	Recoveries this month	■3,50,000
NetLoss_ThisMonth	DECIMAL(15,2)	Net loss this month	■8,50,000
Prepayments_ThisMonth	DECIMAL(15,2)	Voluntary prepayments	■1,25,00,000
ScheduledAmort	DECIMAL(15,2)	Scheduled principal repayment	■85,00,000
EOP_Balance	DECIMAL(15,2)	Pool balance at end of month	■30,22,00,000
CollectionEfficiency	DECIMAL(5,4)	Collections / Billing	0.9720

6.3.2 Core Dynamic Loss DAX Measures

```
Monthly Default Rate (MDR): VAR BOP_Balance = CALCULATE(SUM(DynamicLoss[BOP_Balance]),  
DimDate[Date] = MAX(DimDate[Date])) VAR NewDefaults =  
CALCULATE(SUM(DynamicLoss[NewDefaults_Balance]), DimDate[Date] = MAX(DimDate[Date])) RETURN  
DIVIDE(NewDefaults, BOP_Balance) Constant Default Rate (CDR) - Annualised: CDR = 1 - POWER(1 -  
[Monthly Default Rate], 12) Monthly Net Loss Rate: VAR AvgBalance = DIVIDE(  
CALCULATE(SUM(DynamicLoss[BOP_Balance])) + CALCULATE(SUM(DynamicLoss[EOP_Balance])), 2) RETURN  
DIVIDE(CALCULATE(SUM(DynamicLoss[NetLoss_ThisMonth])), AvgBalance) Annualised Net Loss Rate:  
Annualised Net Loss Rate = [Monthly Net Loss Rate] * 12 Single Month Mortality (SMM): VAR BOP =  
CALCULATE(SUM(DynamicLoss[BOP_Balance])) VAR SchedPrin =  
CALCULATE(SUM(DynamicLoss[ScheduledAmort])) VAR Prepay =  
CALCULATE(SUM(DynamicLoss[Prepayments_ThisMonth])) RETURN DIVIDE(Prepay, BOP - SchedPrin)  
Conditional Prepayment Rate (CPR): CPR = 1 - POWER(1 - [SMM], 12) Excess Spread: VAR  
InterestCollected = [Pool Interest Collections] VAR InvestorCouponDue = [Total Tranche Interest  
Due] VAR ServicingFee = [Servicer Fee Amount] VAR TrusteeFee = [Trustee Fee Amount] VAR  
LossesAbsorbed = CALCULATE(SUM(DynamicLoss[NetLoss_ThisMonth])) RETURN InterestCollected -  
InvestorCouponDue - ServicingFee - TrusteeFee - LossesAbsorbed
```

6.4 Roll Rate Analysis

Roll rate analysis tracks how loans migrate between DPD buckets month-over-month. This is the most critical predictive tool for estimating future losses. Rating agencies use roll rates to calibrate their loss models, and investors monitor them for early warning signals. The DAX implementation requires CALCULATETABLE with PREVIOUSMONTH to track loan-level transitions.

6.4.1 Roll Rate Matrix - Standard Format

The roll rate matrix shows the probability of a loan moving from one DPD bucket to another in a single month. This matrix is displayed as a heatmap in Power BI, with color intensity indicating transition probability:

From \ To	Current	1-29 DPD	30-59 DPD	60-89 DPD	90+ DPD	Default
Current	96.5%	2.8%	0.5%	0.1%	0.05%	0.05%
1-29 DPD	45.0%	30.0%	20.0%	3.0%	1.5%	0.5%
30-59 DPD	15.0%	10.0%	35.0%	30.0%	8.0%	2.0%
60-89 DPD	5.0%	3.0%	7.0%	30.0%	40.0%	15.0%
90+ DPD	2.0%	1.0%	2.0%	5.0%	50.0%	40.0%

Interpretation: The matrix reveals that 96.5% of current loans remain current, while 45% of 1-29 DPD loans cure back to current. The critical transition is from 60-89 DPD to 90+ DPD (40%), which represents the 'roll-to-default' pipeline. Early intervention at the 30-59 DPD stage can prevent 30% of loans from rolling to 60-89 DPD.

6.4.2 Roll Rate DAX Implementation

```
Roll Rate from Bucket A to Bucket B: VAR FromBucket = SELECTEDVALUE(RollRateMatrix[FromBucket])
VAR ToBucket = SELECTEDVALUE(RollRateMatrix[ToBucket]) VAR CurrentDate = MAX(DimDate[Date]) VAR
PriorDate = EDATE(CurrentDate, -1) VAR LoansInFromBucket = CALCULATETABLE(
VALUES(FactLoanPerformance[LoanID]), FactLoanPerformance[DPD_Bucket] = FromBucket, DimDate[Date]
= PriorDate) VAR BalanceInFromBucket = CALCULATE( SUM(FactLoanPerformance[CurrentBalance]),
FactLoanPerformance[DPD_Bucket] = FromBucket, DimDate[Date] = PriorDate) VAR BalanceTransitioned
= CALCULATE( SUM(FactLoanPerformance[CurrentBalance]), FactLoanPerformance[DPD_Bucket] =
ToBucket, DimDate[Date] = CurrentDate, TREATAS(LoansInFromBucket, FactLoanPerformance[LoanID]))
RETURN DIVIDE(BalanceTransitioned, BalanceInFromBucket) Cure Rate (loans returning to Current):
VAR CurrentDate = MAX(DimDate[Date]) VAR PriorDate = EDATE(CurrentDate, -1) VAR DelinqLastMonth =
CALCULATETABLE( VALUES(FactLoanPerformance[LoanID]), FactLoanPerformance[DPD_Days] >= 30,
DimDate[Date] = PriorDate) VAR DelinqBalanceLastMonth = CALCULATE(
SUM(FactLoanPerformance[CurrentBalance]), FactLoanPerformance[DPD_Days] >= 30, DimDate[Date] =
PriorDate) VAR CuredBalance = CALCULATE( SUM(FactLoanPerformance[CurrentBalance]),
FactLoanPerformance[DPD_Days] = 0, DimDate[Date] = CurrentDate, TREATAS(DelinqLastMonth,
FactLoanPerformance[LoanID])) RETURN DIVIDE(CuredBalance, DelinqBalanceLastMonth) Roll-to-Default
Rate (60-89 DPD to 90+ DPD): VAR CurrentDate = MAX(DimDate[Date]) VAR PriorDate =
EDATE(CurrentDate, -1) VAR Loans60to89 = CALCULATETABLE( VALUES(FactLoanPerformance[LoanID]),
FactLoanPerformance[DPD_Days] >= 60 && FactLoanPerformance[DPD_Days] < 90, DimDate[Date] =
PriorDate) VAR Balance60to89 = CALCULATE( SUM(FactLoanPerformance[CurrentBalance]),
FactLoanPerformance[DPD_Days] >= 60 && FactLoanPerformance[DPD_Days] < 90, DimDate[Date] =
```

```
PriorDate) VAR RolledToDefault = CALCULATE( SUM(FactLoanPerformance[CurrentBalance]),  
FactLoanPerformance[DPD_Days] >= 90, DimDate[Date] = CurrentDate, TREATAS(Loans60to89,  
FactLoanPerformance[LoanID])) RETURN DIVIDE(RolledToDefault, Balance60to89)
```

Key Insights from Roll Rate Analysis:

- **High Cure Probability:** 45% of 1-29 DPD loans cure back to current, supporting early collection intervention strategies
- **Critical Transition Point:** 30-59 DPD to 60-89 DPD transition (30%) is the highest risk pipeline requiring immediate attention
- **Default Cascade:** 40% of 60-89 DPD loans roll to 90+ DPD, and 50% of 90+ DPD loans remain in default, indicating severe delinquency persistence
- **Early Warning Signal:** A decline in cure rates or increase in roll-to-default rates indicates portfolio deterioration requiring stress testing

7. WATERFALL & TRANCHE ALLOCATION

The cash flow waterfall is the structural heart of any securitisation transaction. It defines the priority of payments across tranches and ensures that senior investors receive protection through subordination, excess spread, and reserve accounts. The DAX implementation models the sequential allocation logic, trigger testing, and credit enhancement tracking required for monthly investor reporting and rating agency surveillance.

7.1 Waterfall Structure

The standard cash flow waterfall for an auto loan ABS transaction follows this priority order:

- **1. Total Collections:** Interest + Principal + Prepayments + Recoveries received from servicer
- **2. Less: Senior Fees:** Trustee fee, Servicer fee, Rating agency fee (paid first priority)
- **3. Less: Senior Tranche Interest:** Coupon payment to AAA/AA rated senior note holders
- **4. Less: Senior Tranche Principal:** Scheduled amortization or prepayment allocation to senior tranche
- **5. Less: Mezzanine Tranche Interest:** Paid only if OC/IC tests pass (sequential pay trigger)
- **6. Less: Mezzanine Tranche Principal:** Paid after senior principal in full is satisfied
- **7. Residual to Equity/Originator:** Excess spread released to equity tranche or retained by originator
- **8. Reserve Account:** Target balance maintained for liquidity support and loss absorption

7.2 Trigger and Covenant Testing

Structural triggers protect senior tranche investors by redirecting cash flows when portfolio performance deteriorates. The DAX model implements real-time trigger testing with PASS/FAIL indicators:

Test	Trigger Level	Current Level	Status	Action if Failed
OC Test (Senior)	>= 120%	125%	PASS	Sequential pay if fail
OC Test (Mezz)	>= 110%	115%	PASS	Divert to deleverage
IC Test (Senior)	>= 125%	130%	PASS	Trap excess spread
IC Test (Mezz)	>= 115%	120%	PASS	Redirect cash flows
Delinquency Test	<= 5.0%	0.05%	PASS	Early amortisation
Cumulative Loss Test	<= 3.0%	0.01%	PASS	Accelerate paydown

7.3 DAX Waterfall Implementation

Credit Enhancement Percentage: VAR SubordinationAmt = [Subordinated Tranche Balance] VAR ExcessSpreadReserve = [Reserve Account Balance] VAR OC_Amount = SUM(Loans[CurrentBalance]) - [Senior Tranche Balance] RETURN DIVIDE(SubordinationAmt + ExcessSpreadReserve + OC_Amount, SUM(Loans[CurrentBalance])) **Reserve Account Adequacy:** Reserve Adequacy = DIVIDE([Reserve Account Balance], [3M Avg Monthly Losses]) **OC Test (Senior):** VAR SeniorOC = DIVIDE(SUM(Loans[CurrentBalance]), [Senior Tranche Balance]) RETURN IF(SeniorOC >= 1.20, "PASS", "FAIL") **IC Test (Senior):** VAR SeniorIC = DIVIDE([Collections Total], [Senior Tranche Interest Due]) RETURN IF(SeniorIC >= 1.25, "PASS", "FAIL") **Delinquency Trigger Test:** VAR DelinqRate = [30Plus DPD Rate] RETURN IF(DelinqRate <= 0.05, "PASS", "FAIL") **Cumulative Loss Trigger Test:** VAR CumLossRate = [Cumulative Net Loss Rate] RETURN IF(CumLossRate <= 0.03, "PASS", "FAIL")

Waterfall Health Assessment: All trigger tests currently show PASS status, indicating robust structural protection. The senior OC test at 125% provides 5% cushion above the 120% trigger, while the delinquency test at 0.05% is 100x below the 5% threshold. This structural strength supports the AAA rating assumption for the senior tranche.

8. STRESS TESTING FRAMEWORK

The stress testing framework enables scenario-based risk assessment by applying shock parameters to base case assumptions. This is critical for rating agency stress scenarios (S&P; LEVELS: 3.0x-4.0x base loss for AAA; Moody's: 2.5x-3.0x for AA), regulatory capital adequacy assessment, and investor due diligence. The DAX implementation uses What-If parameters for interactive scenario comparison.

8.1 What-If Parameters

The stress testing dashboard incorporates five interactive parameters that users can adjust to model different economic scenarios:

Parameter	Base Case	Moderate Stress	Severe Stress	Crisis
Default Rate Multiplier	1.0x	2.0x	3.5x	5.0x
Recovery Rate Haircut	0%	20%	40%	60%
Prepayment Speed Change	0%	-30%	-50%	-70%
Interest Rate Shock	0 bps	+100 bps	+250 bps	+500 bps
Property Value Decline	0%	-15%	-30%	-50%

8.2 Stress-Adjusted DAX Measures

Stressed Default Rate: $\text{Stressed Default Rate} = [\text{Base Default Rate}] * [\text{Default Rate Multiplier Parameter}]$ **Stressed Recovery Rate:** $\text{Stressed Recovery Rate} = [\text{Base Recovery Rate}] * (1 - [\text{Recovery Rate Haircut Parameter}])$ **Stressed LGD:** $\text{Stressed LGD} = 1 - [\text{Stressed Recovery Rate}]$ **Stressed ECL:** $\text{Stressed ECL} = \text{SUMX}(\text{Loans}, [\text{Stressed PD}] * [\text{Stressed LGD}] * \text{Loans}[\text{EAD}])$ **S&P; Stress Scenario (AAA):** $\text{SP Stressed Loss AAA} = \text{VAR BaseLoss} = [\text{Base Case Expected Loss}]$ $\text{VAR StressMultiple} = 3.5$ // AAA stress factor $\text{VAR StressedLoss} = \text{BaseLoss} * \text{StressMultiple}$ $\text{VAR CreditEnhancement} = [\text{Senior Tranche CE Percentage}]$ $\text{RETURN IF}(\text{StressedLoss} > \text{CreditEnhancement}, \text{"INSUFFICIENT CE"}, \text{"ADEQUATE CE"})$ **Market Value Decline Stressed LTV:** $\text{Stressed LTV} = \text{VAR CurrentLTV} = [\text{Weighted Avg LTV}]$ $\text{VAR MVD_Haircut} = 0.20$ // 20% market value decline $\text{VAR StressedPropertyValue} = \text{SUMX}(\text{Loans}, \text{Loans}[\text{CurrentPropertyValue}] * (1 - \text{MVD_Haircut}))$ $\text{VAR TotalBalance} = \text{SUM}(\text{Loans}[\text{CurrentBalance}])$ $\text{RETURN DIVIDE}(\text{TotalBalance}, \text{StressedPropertyValue})$

8.3 Scenario Comparison Dashboard

The stress testing dashboard provides side-by-side comparison of four scenarios:

Metric	Base Case	Moderate Stress	Severe Stress	Crisis
30+ DPD Rate	0.05%	0.10%	0.18%	0.25%

Metric	Base Case	Moderate Stress	Severe Stress	Crisis
CDR	0.04%	0.08%	0.14%	0.20%
Annualised Net Loss Rate	0.01%	0.03%	0.06%	0.10%
Recovery Rate	36%	29%	22%	14%
Excess Spread (Monthly)	■18.04M	■16.20M	■12.50M	■8.00M
ECL Provision	■2.5M	■7.8M	■18.5M	■35.2M
Senior Tranche CE Coverage	4.2x	3.1x	2.0x	1.2x
Trigger Status	ALL PASS	ALL PASS	OC Test FAIL	Multiple FAIL

Stress Test Conclusion: The portfolio demonstrates strong resilience under moderate stress scenarios, with all triggers remaining in PASS status. Under severe stress (3.5x default multiplier), the OC Test for senior tranche fails at 2.0x coverage, triggering sequential pay mechanism. In crisis scenarios, multiple trigger breaches occur, requiring early amortisation and accelerated paydown. The current excess spread of ■216.53M provides substantial cushion, equivalent to 999+ months of current monthly losses.

9. INVESTOR REPORTING & REGULATORY COMPLIANCE

Institutional investors (pension funds, insurance companies, mutual funds, hedge funds) require standardized monthly reporting from the servicer/trustee. The Power BI dashboard generates these reports with full DAX automation, ensuring consistency, timeliness, and auditability. This section covers the six standard sections of the monthly servicer report, rating agency data tape requirements, and regulatory compliance with SEBI and RBI frameworks.

9.1 Monthly Servicer Report

The monthly servicer report is the primary communication channel between the servicer, trustee, and investors. It must be produced within T+15 business days of the reporting period end. The DAX model automates all six standard sections:

9.1.1 Section 1: Pool Summary

- **Beginning-of-Period Pool Balance:** Opening balance at start of reporting month
- **Scheduled Principal Collected:** Regular amortization payments received
- **Unscheduled Principal (Prepayments):** Voluntary prepayments and refinancings
- **Losses Realised:** Gross losses from defaulted loans
- **Recoveries on Prior Losses:** Collections from previously written-off loans
- **End-of-Period Pool Balance:** Closing balance after all adjustments
- **Pool Factor:** Current Balance / Original Balance ratio
- **Number of Loans:** Beginning, Added, Removed, and Ending counts

9.1.2 Section 2: Delinquency and Default Report

Delinquency Bucket	# Loans	Balance (INR)	% of Pool	Migration
Current	385	■317.79M	77.0%	Cured from 30 DPD
1-29 DPD	37	■15.20M	7.4%	New delinquencies
30-59 DPD	14	■8.50M	2.8%	Rolled from 1-29
60-89 DPD	8	■4.80M	1.2%	Rolled from 30-59
90-119 DPD	5	■3.20M	0.8%	Rolled from 60-89
120+ DPD	3	■2.10M	0.5%	Rolled from 90-119
Default	2	■1.50M	0.4%	New defaults

Delinquency Bucket	# Loans	Balance (INR)	% of Pool	Migration
Reposessed	1	■0.80M	0.2%	In recovery

9.1.3 Section 3: Loss and Recovery Report

- **Cumulative Gross Losses:** Total realised losses since inception
- **Cumulative Recoveries:** Total collections from defaulted loans
- **Cumulative Net Losses:** Gross Losses - Recoveries
- **Annualised Net Loss Rate:** (Monthly Net Loss / Avg Balance) × 12
- **Recovery Rate:** Recoveries / Gross Losses
- **Average Time to Recovery:** Months from default to recovery (typically 3-6 months for auto)

9.1.4 Section 4: Prepayment Report

- **Monthly Prepayment Amount:** Total voluntary prepayments
- **Single Month Mortality (SMM):** Prepay / (BOP - Scheduled Principal)
- **Conditional Prepayment Rate (CPR):** Annualized: $1 - (1 - \text{SMM})^{12}$
- **Constant Prepayment Rate by Vintage:** CPR segmented by origination quarter

9.1.5 Section 5: Cash Flow Waterfall Report

- **Total Collections:** Interest + Principal + Prepayments + Recoveries
- **Less: Senior Fees:** Trustee, Servicer, Rating Agency fees
- **Less: Senior Tranche Interest:** Coupon payment to senior note holders
- **Less: Senior Tranche Principal:** Scheduled amortization or prepayment allocation
- **Less: Mezzanine Tranche Interest:** Paid if OC/IC tests pass
- **Less: Mezzanine Tranche Principal:** Paid after senior principal in full
- **Residual to Equity/Originator:** Excess spread released
- **Reserve Account Balance:** Target balance maintained for liquidity support

9.1.6 Section 6: Trigger/Covenant Status

Test	Trigger Level	Current Level	Status	Action Required
OC Test (Senior)	>= 120%	125%	PASS	None
OC Test (Mezz)	>= 110%	115%	PASS	None
IC Test (Senior)	>= 125%	130%	PASS	None

Test	Trigger Level	Current Level	Status	Action Required
IC Test (Mezz)	>= 115%	120%	PASS	None
Delinquency Test	<= 5.0%	0.05%	PASS	None
Cumulative Loss Test	<= 3.0%	0.01%	PASS	None

9.2 Rating Agency Data Tapes

Rating agencies require loan-level data tapes for initial rating and ongoing surveillance. The Power BI model generates these tapes on-demand with all mandatory fields populated through DAX measures. The following tables outline the requirements for S&P, Moody's, and Indian agencies (CRISIL/ICRA):

9.2.1 S&P; Global LEVELS Model Requirements

- **Base Case Loss Expectation:** Derived from historical loss curves of the originator
- **Loss Timing Curve:** Monthly distribution of expected losses (peaks 18-30 months for auto)
- **Recovery Lag:** Average time from default to recovery (typically 3-6 months for auto)
- **Voluntary Prepayment Speed:** Expressed as ABS or CPR
- **Recovery Rate by LTV Band:** Different recovery assumptions for high-LTV vs low-LTV loans
- **Stress Multiples by Rating Level:** AAA = 3.0x-4.0x base loss, AA = 2.5x-3.0x, A = 2.0x-2.5x

9.2.2 Moody's MILAN Framework Requirements

- **Lognormal Loss Distribution Parameters:** Mean and standard deviation of loss
- **Correlation Assumptions:** Intra-pool asset correlation (typically 3-8% for auto)
- **Seasoning Adjustment:** Performance credit for seasoned pools
- **Originator Assessment:** Qualitative score of originator's underwriting standards
- **Servicer Quality Assessment:** Score of servicer's collection capabilities

9.2.3 Indian Rating Agencies (CRISIL/ICRA) Requirements

- **RBI Priority Sector Classification:** Whether loans qualify under PSL guidelines
- **GST Registration of Borrower:** For commercial vehicle loans
- **Aadhaar Verification Status:** KYC completeness indicator
- **NACH Mandate Status:** Auto-debit consent for EMI collection
- **Insurance Coverage:** Comprehensive vs third-party vs uninsured
- **Hypothecation Registration:** RC book endorsement status
- **Loan Restructuring Under RBI Moratorium:** COVID moratorium status history

9.3 SEBI & RBI Compliance

The Reserve Bank of India issued its revised framework for securitisation of standard assets in September 2021 (RBI/2021-22/85), replacing the 2006 guidelines. The revised framework introduced significant changes

that impact data analytics requirements:

Provision	Requirement	DAX Analytics Impact
Minimum Retention (MRR)	10% of each tranche	Track retained portion per tranche; compliance dashboard
Minimum Holding Period	Based on repayment frequency	Date-based eligibility measure using DATEDIFF
Structure Types	Securitisation or Direct Assignment only	Pool classification and reporting segregation
Disclosure Requirements	Enhanced originator and investor reporting	Automated loan-level and pool-level reporting measures
Credit Enhancement	Must be disclosed and tracked	CE computation measures; tranche subordination tracking
Skin-in-the-Game	First loss retention	Retained interest valuation and risk-weighting measures

SEBI Disclosure Requirements: For Indian securitisations, SEBI requires issuers to provide investor data through designated portals. The following reporting frequencies apply:

Report	Frequency	Deadline	Recipient
Loan-Level Data Tape	Monthly	T+15 business days	Rating Agencies, Investors
Servicer Report	Monthly	T+15 business days	Trustee, Investors
Pool Performance Report	Monthly	T+15 business days	SEBI, Stock Exchange
Compliance Certificate	Quarterly	T+30 calendar days	Trustee, Rating Agencies
Annual Review	Annually	Within 6 months of FY end	Rating Agencies
Material Event Notice	As required	Within 2 business days	All parties

10. CASE STUDIES APPLICATION

The securitisation analytics framework is applied to three major real-world cases to demonstrate its practical utility in crisis detection, regulatory response, and risk mitigation. These case studies validate the DAX measures and dashboard designs against historical events, ensuring that the analytics platform can detect early warning signals and support informed decision-making.

10.1 The 2008 Global Financial Crisis - Securitisation Meltdown

The 2007–2009 Global Financial Crisis was, at its core, a securitisation crisis. US subprime mortgage-backed securities, which had been rated AAA by major rating agencies, experienced catastrophic default rates as the housing bubble burst. The crisis revealed fundamental flaws in securitisation analytics: models assumed perpetually rising house prices, correlation assumptions between mortgage defaults were grossly underestimated, and loan-level data quality was compromised by originate-to-distribute incentive misalignment.

10.1.1 Analytics Failures and DAX Solutions

Lesson	Analytics Failure	DAX Solution
Vintage Analysis	Performance not tracked by origination cohort; deterioration diluted in pool averages	CALCULATE with FILTER on origination date; DATESYTD for cumulative loss curves
Roll Rate Early Warning	Transition from 30 DPD to 60 DPD not monitored systematically	CALCULATE with PREVIOUSMONTH; automated roll rate matrix and alerts
Concentration Risk	Geographic concentration in CA, FL, NV, AZ amplified losses	HHI measure using SUMX; TOPN exposure reporting; geographic heatmaps
Stress Testing Severity	Tests used mild 5-10% decline; actual exceeded 30%	What-if parameters in Power BI; multiple severity scenario measures

10.1.2 Application to Current Project

The current project's DAX implementation addresses all four failure points identified in the 2008 crisis. The vintage analysis dashboard (Screenshot 20) tracks performance by origination cohort, preventing dilution of deteriorating vintages in pool averages. The roll rate matrix (Section 6.4) provides automated early warning detection for delinquency migration. The HHI concentration measure (Section 4.3) identifies geographic and borrower concentrations before they amplify losses. The stress testing framework (Section 8) models severe scenarios up to 5.0x base loss with 60% recovery haircuts, far exceeding the mild assumptions that failed in 2008.

10.2 The IL&FS; Crisis (2018) - Indian Structured Finance Collapse

In September 2018, Infrastructure Leasing & Financial Services (IL&FS;), one of India's largest infrastructure finance companies, defaulted on its commercial paper and bond obligations, triggering a crisis across India's structured finance and NBFC sectors. IL&FS; group entities had issued securitised instruments backed by infrastructure project receivables and loan portfolios. The crisis exposed cascading risks: SPV cash flows were disrupted as underlying project payments stalled, investors in pass-through certificates (PTCs) faced payment delays, and the broader NBFC sector experienced a liquidity crunch.

10.2.1 Data Analytics Implications

- **Counterparty Concentration:** Several securitisation pools had exposure to IL&FS; group entities as borrowers or guarantors. The DAX measures for obligor concentration (top 10 borrower exposure, single-name concentration limits) are critical for detecting such risks.
- **Real-Time Collection Monitoring:** The IL&FS; crisis demonstrated that monthly reporting cycles are insufficient for detecting cash flow disruptions. DAX time-intelligence measures with daily or weekly granularity, combined with collection efficiency ratios, provide the real-time monitoring that was absent.
- **RBI Regulatory Response:** Post-IL&FS;, the RBI revised its securitisation framework (September 2021), introducing stricter Minimum Holding Period and Minimum Retention Requirements. The revised framework requires originators to retain a minimum 10% of each tranche issued, ensuring skin-in-the-game.

10.3 RBI Revised Securitisation Framework (2021)

The Reserve Bank of India issued its revised framework for securitisation of standard assets in September 2021 (RBI/2021-22/85 DOR.STR.REC.51/21.04.177/2021-22), replacing the 2006 guidelines. The revised framework introduced significant changes that directly impact the data analytics requirements for this project:

- **Minimum Retention Requirement (MRR):** 10% of each tranche must be retained by the originator, ensuring alignment of interests
- **Minimum Holding Period (MHP):** Based on repayment frequency, ensuring loans have sufficient seasoning before securitisation
- **Simplified Structures:** Only two categories allowed—securitisation and direct assignment—reducing structural complexity
- **Enhanced Disclosure:** Originators must provide granular loan-level data and pool-level performance summaries
- **Credit Enhancement Disclosure:** All forms of CE must be disclosed and tracked monthly
- **Skin-in-the-Game:** First loss retention requirements ensure originators bear initial risk

Impact on DAX Implementation: The revised framework requires enhanced data analytics capabilities. The DAX measures must track MRR compliance (retained portion per tranche), generate standardized pool reports matching RBI specifications, compute retained interest valuation for risk-weighting, and produce monthly compliance certificates. The current project's 80+ measure library includes specific measures for NACH mandate coverage, insurance coverage rate, and GST registration tracking—all required by the revised framework.

11. GAMIFIED SIMULATION: SECURITISATION RISK ARENA

The Securitisation Risk Arena is a browser-based gamified simulation platform designed to train and assess data analysts in the specialised discipline of securitisation risk assessment through realistic, time-pressured, scenario-driven challenges using advanced DAX analytics. The platform addresses the critical training gap in structured finance analytics by providing a zero-risk environment where learners experience every possible analytics scenario—from routine monthly reporting to crisis-mode stress testing.

11.1 Business Rationale and Problem Statement

Traditional securitisation analytics training relies on passive reading of rating agency methodologies and occasional shadowing of senior analysts. The consequences of errors in securitisation analytics are severe: an incorrect ECL computation can misstate the bank's provision requirements by millions; a flawed waterfall calculation can misallocate cash flows between senior and subordinated tranches; a missed delinquency trend can delay early-warning triggers, leading to rating downgrades and investor redemptions. India's securitisation market has grown significantly following the RBI's revised framework, with issuance volumes exceeding **■1.8** lakh crore annually. Yet there is no standardised, scalable training platform for securitisation data analysts that combines DAX proficiency with domain expertise.

11.2 Simulation Architecture

Component	Description	Business Value
Portfolio Library	Pre-built datasets for 6 fictional securitisation pools (500+ loans per pool)	Realistic field structures matching rating agency requirements
Market Simulator	Monthly data generator simulating delinquency, prepayment, loss, and recovery	Parameterised economic scenarios: benign, moderate stress, severe stress, crisis
DAX Challenge Engine	Specific analytical tasks requiring DAX measure construction	From basic aggregations to complex iterator patterns and virtual tables
Waterfall Engine	Simulated cash flow waterfall with sequential allocation	Errors cascade through tranche payments, revealing mistakes immediately
Rating Agency Module	Loan-level data tape generation for S&P, Moody's, CRISIL, ICRA	Complete validation checks and compliance testing
ECL Engine	Embedded IFRS 9 computation module	Evaluates PD, LGD, EAD measures for stage classification
Scoring Engine	Awards points for accuracy, performance, completeness, and insight	Deducts points for errors, slow queries, and missed risk signals

11.3 1000-Point Scoring Framework

Scoring Category	Points	Key Criteria
DAX Accuracy & Complexity	300	All measures correct; appropriate patterns; proper context transition
Query Performance	150	All measures <100ms in DAX Studio; efficient variables and filter propagation
Waterfall & Tranche Allocation	200	Complete waterfall with correct sequential allocation, trigger testing, CE tracking
IFRS 9 ECL Computation	150	ECL matches Excel crosscheck within 0.01% variance; correct stage classification
Investor Reporting Completeness	100	All 6 monthly servicer report sections automated with correct DAX measures
Dashboard Quality & Business Insight	100	8+ interactive pages; drill-through; RLS for 5+ roles; business insights narrative
TOTAL	1000	Complete mastery across all dimensions

11.4 5-Year Revenue Projection

Year	B2B SaaS	Retail	University	Certification	Total
Year 1	■6 lakh	■10 lakh	■16 lakh	-	≈■32 lakh
Year 2	■45 lakh	■25 lakh	■50 lakh	-	≈■1.2 crore
Year 3	■1.5 crore	■1 crore	■1.5 crore	■20 lakh	≈■4.2 crore
Year 4	■4 crore	■2.5 crore	■3 crore	■80 lakh	≈■10.3 crore
Year 5	■8 crore	■5 crore	■5 crore	■2 crore	≈■20 crore

12. CONCLUSION & RECOMMENDATIONS

This project demonstrates the successful development of a comprehensive securitisation risk assessment platform using Microsoft Power BI and advanced DAX analytics. The five-dashboard suite provides end-to-end coverage of portfolio monitoring, from executive-level KPIs to loan-level detail, from current snapshot analysis to vintage cohort tracking, and from base-case projections to stress-tested scenarios. The platform meets rating agency requirements, regulatory compliance standards, and investor reporting expectations.

12.1 Key Achievements

- **Comprehensive Dashboard Suite:** Five interconnected dashboards covering Executive Overview, DPD Analysis, Dynamic Loss, Static Pool, and Risk Monitoring
- **Production-Grade DAX Library:** 80+ measures implementing complex patterns including CALCULATE with filter modifications, SUMX iterators, virtual tables, and time intelligence
- **Regulatory Compliance:** Full alignment with RBI Revised Securitisation Framework (2021) and SEBI disclosure requirements
- **Rating Agency Ready:** Loan-level data tapes, pool stratification tables, and stress scenarios matching S&P; LEVELS, Moody's MILAN, and CRISIL/ICRA specifications
- **Stress Testing Capability:** Interactive what-if parameters enabling scenario comparison from base case to crisis mode
- **Gamified Training Platform:** Securitisation Risk Arena concept with 8 progressive levels, 1000-point scoring, and 5-year revenue projection

12.2 Recommendations for Future Development

- **Complete Risk Monitoring Dashboard:** Resolve placeholder values (1.00) and incomplete measures (--) in KPI cards to ensure full operational readiness
- **Implement Row-Level Security (RLS):** Configure role-based access for Portfolio Manager, Risk Analyst, Senior Management, Auditor, and Servicer roles
- **Performance Optimization:** Test all measures in DAX Studio with <100ms target; optimize using variables and reduce context transitions
- **Automated Data Pipeline:** Build straight-through processing from SAP/ERP systems to Power BI model for real-time portfolio updates
- **Enhanced ESG Metrics:** Add environmental (EV percentage), social (affordable housing), and governance (compliance score) measures as emerging standards
- **Predictive Analytics:** Integrate machine learning models for PD forecasting and early default prediction using Python/R scripts in Power BI
- **Mobile Responsiveness:** Optimize dashboard layouts for tablet and smartphone viewing for executive on-the-go access

- **Alert System:** Implement automated email/SMS alerts when triggers approach breach thresholds (e.g., 80% of trigger level)

12.3 Business Impact

The securitisation analytics platform delivers measurable business value across multiple dimensions. For originators, it reduces reporting time from days to hours through DAX automation. For investors, it provides transparent, auditable risk metrics that support informed investment decisions. For regulators, it ensures compliance with RBI and SEBI requirements through standardized reporting. For rating agencies, it produces loan-level data tapes and stress scenarios that streamline surveillance reviews. The platform's gamified training component (Securitisation Risk Arena) addresses the critical talent gap in structured finance analytics, creating a scalable pipeline of skilled data analysts for India's growing securitisation market.

Final Assessment: The project successfully transforms raw loan-level data from SAP/ERP systems into actionable risk intelligence through advanced DAX patterns. The dashboard suite provides the comprehensive risk metrics that credit committees, rating agencies, investors, and regulators demand. With further development of the Risk Monitoring Dashboard and implementation of RLS, the platform will be production-ready for deployment in live securitisation operations.

13.3 Technical References

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13.4 Appendix A: Complete DAX Measure Library

The following table provides a complete index of all 80+ DAX measures developed for this project, organized by functional category:

Category	Measure Name	Business Purpose	Complexity
Pool Health	Total Outstanding Balance	Aggregate portfolio size	Basic
Pool Health	Pool Factor	Amortization progress indicator	Basic
Pool Health	Delinquency Rate	30+ DPD percentage	Basic
Pool Health	Serious Delinquency Rate	60+ DPD percentage	Basic
Pool Health	Collection Efficiency	Actual vs scheduled collections	Basic
Pool Health	CDR	Annualized default rate	Intermediate
Weighted Avg	WAC	Weighted average coupon	Intermediate
Weighted Avg	WAM	Weighted average maturity	Intermediate
Weighted Avg	WALA	Weighted average loan age	Intermediate
Weighted Avg	Weighted Avg LTV	Collateral coverage ratio	Intermediate
Concentration	HHI Index	Portfolio diversification measure	Advanced
Concentration	Top N Borrower Exposure	Single-name concentration risk	Advanced
Concentration	Geographic Concentration	Regional risk exposure	Advanced
Time Intel	YTD Delinquency Rate	Year-to-date aggregation	Intermediate
Time Intel	3M Avg 30Plus DPD Rate	Rolling 3-month average	Intermediate
Time Intel	DPD Trend Direction	Improving/Stable/Deteriorating	Advanced
ECL	IFRS 9 Stage	Stage 1/2/3 classification	Intermediate
ECL	12M ECL	Expected credit loss Stage 1	Advanced

Category	Measure Name	Business Purpose	Complexity
ECL	Lifetime ECL	Expected credit loss Stage 2&3	Advanced
ECL	ECL Coverage Ratio	Provision adequacy	Intermediate
Waterfall	Credit Enhancement %	Total CE calculation	Advanced
Waterfall	Reserve Adequacy	Months of loss coverage	Intermediate
Waterfall	OC Test (Senior)	Overcollateralization test	Intermediate
Waterfall	IC Test (Senior)	Interest coverage test	Intermediate
Static Pool	Static CNL Rate	Cumulative net loss by vintage	Advanced
Static Pool	Projected Ultimate CNL	Loss curve extrapolation	Advanced
Static Pool	Loss Curve Acceleration	Early warning indicator	Advanced
Dynamic Loss	Monthly Default Rate	Point-in-time default velocity	Intermediate
Dynamic Loss	SMM	Single month mortality	Intermediate
Dynamic Loss	CPR	Conditional prepayment rate	Intermediate
Dynamic Loss	Excess Spread	Net interest income cushion	Advanced
Roll Rate	Roll Rate Matrix	Transition probability heatmap	Advanced
Roll Rate	Cure Rate	Loans returning to current	Advanced
Roll Rate	Roll-to-Default Rate	60-89 to 90+ DPD transition	Advanced
Stress Test	Stressed Default Rate	Shock-adjusted default	Advanced
Stress Test	Stressed ECL	Scenario-based provision	Advanced
Stress Test	SP Stressed Loss AAA	Rating agency stress test	Advanced

13.5 Appendix B: Screenshot Reference Guide

Screenshot	Dashboard	Page	Key Visuals	Status
Screenshot (16)	Executive Overview	Page 1 (Design Mode)	Donut, Bar charts, Map, KPI cards	Complete
Screenshot (17)	Executive Overview	Page 1 (View Mode)	Same as above, all tabs visible	Complete
Screenshot (18)	DPD Analysis	Page 2	KPI cards, DPD distribution, trend, gauge	Complete
Screenshot (19)	Dynamic Loss	Page 3	KPI cards, waterfall, collections, scatter	Complete
Screenshot (20)	Static Pool	Page 4	KPI cards, loss curves, vintage table, slicers	Complete

Screenshot	Dashboard	Page	Key Visuals	Status
Screenshot (21)	Risk Monitoring	Page 5	KPI cards, region chart, detail table, gauge	Partial (placeholders)

13.6 Appendix C: 15-Day Project Timeline

Day	Focus Area	Key Deliverable
Day 1	Domain Orientation	Securitisation landscape summary with regulatory framework map
Day 2	Data Architecture & Star Schema	Power BI model with star schema and relationships
Day 3	Core DAX Measures (30+)	Measure dictionary; pool balance, count, and weighted average measures
Day 4	Time Intelligence & Trends	YTD/QTd/MTD/rolling measures; trend analysis dashboard
Day 5	Advanced DAX & Virtual Tables	Stratification measures; concentration analysis; HHI
Day 6	Waterfall & Tranche Allocation	Waterfall model; trigger testing; credit enhancement tracking
Day 7	DPD & Roll Rate Analysis	DPD dashboard; roll rate matrix; ageing report
Day 8	Vintage/Static Pool Analysis	Cumulative loss curves; vintage comparison charts
Day 9	IFRS 9 ECL Framework	ECL model; stage classification; Excel crosscheck
Day 10	Stress Testing & What-If	Stress testing dashboard with interactive parameters
Day 11	Investor Reporting & Data Tapes	6 servicer report sections; rating agency data tape export
Day 12	Security, Governance & Testing	RLS for 5+ roles; DAX Studio performance report
Day 13	Case Studies & Documentation	First-draft report; two case study analyses
Day 14	Validation & Polish	Validated draft; optimised model; presentation deck
Day 15	Final Compilation & Handover	Final PDF + presentation + model + Excel + GitHub transfer

INTELLECTUAL PROPERTY PROTECTION PROTOCOL

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